SPECIALIZATION LIFTING ALONG ALL BASE CHANGES EQUALS THE VALUATIVE CRITERION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4069-4126, following the statement at 4058-4067.

Editorial note. The base-change half of (2)=>(1) is a diagram-anchored construction reading ('It is immediate ... by definition of the fibre product', ~4096-4108); reproduced in the notes, same precedent as cards L26341/L26356.

Allowed standard prerequisites (editorial).

- Points and residue field extensions. lemma-characterize-points (same chapter, cited at ~4077-4078 and the discussion following). Gives kappa(x') subset K and the local ring map O_{X,x} -> kappa(x').
- Specializations give local ring maps. lemma-specialize-points (same chapter, cited at ~4083). Supplies the local map A -> O_{X,x} from the chosen specialization.
- Domination characterization of valuation rings. Algebra, Definition algebra-definition-valuation-ring (cross-chapter, cited at ~4089). Forces the image of O_{X,x} -> K to equal A.
- Morphism from Spec of a local ring. lemma-morphism-from-spec-local-ring (same chapter, cited at ~4092). Turns the ring map O_{X,x} -> A into Spec(A) -> X.
- Specializations and field extensions. lemma-points-specialize (same chapter, cited at ~4111). Produces the commutative diagram for the converse direction.

Source reference. Original reference 00I9: focused-seeds/geometry/S9-schemes-lift-specializations-valuative/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 01KD: sources/stacks-schemes/pinned-original-source.tex, source lines 3995–3995. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01J9: sources/stacks-schemes/pinned-original-source.tex, source lines 2682–2682. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof

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Source reference. Original lemma 01J6: sources/stacks-schemes/pinned-original-source.tex, source lines 2550–2550. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01J8: sources/stacks-schemes/pinned-original-source.tex, source lines 4025–4025. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01J7: sources/stacks-schemes/pinned-original-source.tex, source lines 2612–2612. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. The following are equivalent*

(1) *Specializations lift along any base change of f*
(2) *The morphism f satisfies the existence part of the valuative criterion.*

Proof. Assume (1) holds. Let a solid diagram as in Definition 01KD be given. In order to find the dotted arrow we may replace $X \to S$ by $X_{\mathrm{Spec}(A)} \to \mathrm{Spec}(A)$ since after all the assumption is stable under base change. Thus we may assume $S = \mathrm{Spec}(A)$. Let $x' \in X$ be the image of $\mathrm{Spec}(K) \to X$, so that we have $\kappa(x') \subset K$, see Lemma 01J9. By assumption there exists a specialization $x' \rightsquigarrow x$ in X such that x maps to the closed point of $S = \mathrm{Spec}(A)$. We get a local ring map $A \to \mathcal{O}_{X,x}$ and a ring map $\mathcal{O}_{X,x} \to \kappa(x')$, see Lemma 01J7 and the discussion following Lemma 01J9. The composition $A \to \mathcal{O}_{X,x} \to \kappa(x') \to K$ is the given injection $A \to K$. Since $A \to \mathcal{O}_{X,x}$ is local, the image of $\mathcal{O}_{X,x} \to K$ dominates A and hence is equal to A, by Algebra, Definition 00I9. Thus we obtain a ring map $\mathcal{O}_{X,x} \to A$ and hence a morphism $\mathrm{Spec}(A) \to X$ (see Lemma 01J6 and discussion following it). This proves (2).

Conversely, assume (2) holds. It is immediate that the existence part of the valuative criterion holds for any base change $X_{S'} \to S'$ of f by considering the following commutative diagram

$$\begin{array}{ccccc} \mathrm{Spec}(K) & \longrightarrow & X_{S'} & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow & & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & S' & \longrightarrow & S \end{array}$$

Namely, the more horizontal dotted arrow will lead to the other one by definition of the fibre product. OK, so it clearly suffices to show that specializations lift along f. Let $s' \rightsquigarrow s$ be a specialization in S, and let $x' \in X$ be a point lying over s'.

Apply Lemma 01J8 to $s' \rightsquigarrow s$ and the extension of fields $K = \kappa(x')/\kappa(s')$. We get a commutative diagram

$$\begin{array}{ccccc} \mathrm{Spec}(K) & & \longrightarrow & & X \\ \downarrow & & & \nearrow & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & \mathrm{Spec}(\mathcal{O}_{S,s}) & \longrightarrow & S \end{array}$$

and by condition (2) we get the dotted arrow. The image x of the closed point of $\mathrm{Spec}(A)$ in X will be a solution to our problem, i.e., x is a specialization of x' and maps to s. □